

Sandbar Volume Change Workflow

A practical QGIS workflow for calculating how sandbar cubic metre volume changes between survey rounds, using repeat elevation surfaces rather than aerial imagery alone.

Core Principle

The most defensible way to measure sandbar change is to compare one survey elevation surface against another inside a fixed polygon. The aerial imagery helps you see change, but the volume calculation itself should be based on elevation difference.

Use for calculation

Elevation surface such as a DSM, or a cleaned bare-earth style surface if available.

Use for interpretation

Aerial imagery, contour views, and section profiles to explain what changed and where.

Recommended QGIS Workflow

1. Prepare each survey surface.

Bring both survey rasters into QGIS and make sure they share the same CRS, resolution, and grid alignment.

2. Create fixed sandbar polygons.

Draw one polygon per sandbar or monitoring zone and keep these polygons unchanged through future surveys.

3. Mask unreliable areas.

Exclude open water, wet reflective zones, noisy edges, vegetation, boats, and any areas you do not trust.

4. Create a difference raster.

"survey_round_2_dsm@1" - "survey_round_1_dsm@1"

Positive values show build-up. Negative values show erosion or lowering.

5. Clip or analyse the difference raster inside each sandbar polygon.

Use clip-by-mask, zonal statistics, or raster-analysis tools depending on the workflow you prefer.

6. Convert height change to volume.

For each cell, multiply height change by cell area. Then sum those values inside the polygon.

7. Report the result for each polygon.

Keep three numbers for every sandbar: gain in cubic metres, loss in cubic metres, and net change in cubic metres.

Volume Logic

Per-cell calculation

$\text{height change} \times \text{cell area} = \text{volume change}$

Polygon totals

$\text{Positive cells} = \text{Gain (m}^3\text{)}$

$\text{Negative cells} = \text{Loss (m}^3\text{)}$

$\text{All cells together} = \text{Net change (m}^3\text{)}$

Best Practice Notes

Keep survey timing comparable

Use similar tidal states whenever possible so the exposed ground is comparable between rounds.

Keep polygons fixed

Do not redraw the sandbar boundaries every survey if your aim is to measure change through time.

Document exclusions

Keep notes on where water, noise, poor overlap, or processing artefacts were excluded from analysis.

Use the app for reporting

Do the calculation in QGIS, then feed the final gain, loss, and net values into the monitoring app.

Suggested Output for the App

The monitoring system can present the volume-change results clearly if each survey comparison exports a small table like this:

```
comparison_id,area_id,sandbar_id,gain_m3,loss_m3,net_m3,notes
2026-03_vs_2026-04,area3,bar_01,128.4,-42.1,86.3,Strong build-up on central ridge
2026-03_vs_2026-04,area5,bar_02,54.7,-97.9,-43.2,Net erosion along seaward edge
```

Recommended next step: calculate the numbers in QGIS first, then bring the results into the app as a survey-comparison dataset.